

Quantity and quality of job creation in renewable energy and energy efficiency: A review of international evidence

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Abstract

In this paper we present findings from a systematic review on job creation, quality, and skills, focusing on decarbonisation in the energy sector. We compare a range of gross job employment factors which indicate that overall, investment in renewable energy and energy efficiency can deliver more jobs than gas or coal power generation. In addition, we review a subset of recent studies which estimate the net employment effects of decarbonisation in the energy sector at a national scale, across various international contexts. These national studies largely agree that the most likely outcome over the next few decades is a modest net positive creation of jobs and moderate economic growth. In certain regions within these countries, jobs in fossil fuel industries may be lost faster than the pace at which low carbon energy sectors can offer new employment. There may be mismatches between regions where displaced workers live and where new opportunities become available, which may be a barrier to accepting alternative employment even if former workers have the requisite skills. In these cases national government transition plans are recommended, co-ordinated with local governments, to manage the impacts of displacement from carbon intensive sectors and respond to the need to build a new low carbon workforce including through skills development and training. We highlight a lack of metrics and data in the literature on job quality, skills, and the geographic distribution of employment impacts in decarbonising energy systems, and these should be priority areas for further research.

1. Introduction

There is a growing international momentum behind the setting of net zero emissions targets in law or policy (Climate Watch, 2023), and the introduction of Green New Deals involving substantial investments in green jobs and infrastructure, for example in the US and EU (Green, 2022). Over the last decade, the costs of leading renewable energy technologies such as solar photovoltaics (PV) and wind power have fallen rapidly or substantially (IRENA, 2020; Jansen *et al.*, 2020). In 2021, there were at least 12.7 million people employed in renewable energy sectors worldwide, both directly and in wider supply chains, with the majority of these jobs concentrated in China, Brazil, the EU, the US and India (IRENA and ILO, 2022). Given the economic implications of the COVID-19 pandemic and recent gas and electricity price spikes (Fernández Alvarez and Molnar, 2021), a key question is whether investment in the low carbon transition can contribute to economic growth and resilience both in the short-term and longer-term (Figueres and Zycher, 2020). There have been various calls for investment in green jobs, skills and infrastructure to help consolidate economic recovery from COVID-19, in a way that is compatible with achieving net zero emissions and a societally just transition (Allan *et al.*, 2020; EEIG, 2020; IEA, 2020; Jung and Murphy, 2020; IRENA and ILO, 2021).

There are various definitions of green jobs, and while it is important to clarify the scope of the present paper in relation to these, we do not set out to contribute further to the existing definitional debate. In the UK for example, the Green Jobs Taskforce consider a green job as “*employment in an activity that directly contributes to - or indirectly supports - the achievement of the UK's net zero emissions target and other environmental goals, such as nature restoration and mitigation against climate risks*” (Green Jobs Taskforce, 2021, p. 15). Green jobs may vary in ‘greenness’, for example in terms of the proportion of tasks carried out on ‘green’ or ‘non-green’ activities (Bowen, Kuralbayeva and Tipoe, 2018). Definitions of green jobs (e.g. International Labour Organization, 2018) may also include a quality aspect, suggesting that green employment needs to be characterised by ‘decent’ work or good quality jobs, e.g. in terms of adequate salaries and safe working conditions (Office for National Statistics, 2021).

A key definitional issue relevant to this paper is the distinction between ‘gross’ and ‘net’ jobs. Gross effects include only the positive impact on employment which may be associated with a particular investment. For any specific energy technology, gross jobs refer to those which are created from project spending on equipment manufacturing, installation, operation, and maintenance (Blyth *et al.*, 2014). Gross jobs can also include employment created in associated supply chains and the wider economy. Net employment is a more holistic metric that accounts for jobs that might be displaced in other parts of the economy as a result of the initial investment. For example, overall net employment could be the number of gross jobs created through additional renewable energy deployment, offset by the implied number of gross jobs that would be lost due to less power generation needed from gas and coal (Ibid.).

Various studies attempt to associate green policies, including fiscal stimuli, financial incentives and regulations, with numbers of jobs created (e.g. Dsouza, 2015; Dvořák *et al.*, 2017; Mundaca and Luth Richter, 2015; Lim, Guzman and Bowen, 2020; Lee, 2017). Dvořák *et al.* (2017) found that job creation in renewable energy in the Czech Republic has depended upon the continuity of financial incentives. A US-focused study suggests that green stimulus programmes supporting wind power and solar PV helped to boost job creation, expand manufacturing capacity and supply chains, and increase revenue from sales of renewable energy technology (Mundaca and Luth Richter, 2015). A review of the impact of various climate change mitigation policies across 20 countries finds that they tend to lead to an overall but modest increase (or otherwise no change) in employment net of jobs displaced (Godinho, 2022). Pre-existing reviews of the literature also suggest that substituting fossil fuels with renewables and energy efficiency is most likely to result in small net employment gains (Blyth *et al.*, 2014; Stavropoulos and Burger, 2020). There is a risk however that some of the newly created jobs may be of poor quality and this may be compounded by insufficient labour market policies or regulations (Godinho, 2022).

In this paper we present findings from a systematic review building upon and updating this existing research. We focus specifically on the energy sector rather than applying a wider definition of green employment. Our paper reviews recent evidence on three aspects of low carbon energy job creation: (1) quantity: how many jobs can be created by policy support for investment in low carbon energy and energy efficiency compared to supporting fossil fuel incumbents? (2) geographic and temporal distribution: how are employment impacts of national decarbonisation strategies likely to be distributed across different regions within countries, and over time? (3) what are the implications of supporting a low carbon energy transition for quality of jobs, skills requirements and training?

Section 2 outlines our approach to addressing these research questions through the systematic review, and includes several observations on the measurement of energy sector job creation. Sections 3 and 4 present the results of the review, which captures a wide range of internationally diverse evidence. In Section 3 we analyse studies which estimate potential quantity of job creation through policy support or investment in different energy production technologies (fossil fuels, renewables, nuclear) and energy efficiency interventions. In Section 4 we consider the extent of identified evidence on quality aspects, occupational characteristics and skills levels of jobs created in fossil fuel, renewable and energy efficiency sectors. Section 5 concludes, suggesting priority areas for policy and further research.

2. Materials And Methods

Our review follows the UKERC Technology and Policy Assessment (TPA) approach (UKERC, 2023) based on the practice of systematic review (Sorrell, 2007). As part of this process, the research approach and outcomes were

informed and critiqued by a small group of expert advisers and policy makers who have brought their experience and perspectives to bear on the research topic. Further details on the TPA process and expert advisers are set out in the Appendix.

2.1. Systematic review approach

The focus of the evidence review is on a low carbon transition in the energy sector, considering particularly (but not exclusively) renewable energy, energy efficiency, end use energy demand sectors, fossil fuels and nuclear power. The geographical coverage is international but limited to evidence available in English. For the systematic review, we selected a range of key words or phrases related to: job creation; climate change and decarbonisation; energy technologies; fuel sources and energy efficiency; policies and financial incentives; quality and skill levels of jobs created and their geographic distribution (Table 1). The search terms were developed based on those used in the 2014 UKERC review on green jobs and our own preliminary, scoping review of the literature. Search terms were combined in search strings and applied to three databases, selected to obtain academic and grey literature across different publishers: Google Scholar, Science Direct and Web of Science. The searches were restricted to the years 2014 to 2022 in order to provide an update to a previous UKERC review on low carbon job creation (Blyth *et al.*, 2014).

Part of the analysis in this study involves comparing low carbon job estimates with estimates of jobs in traditional (fossil-fired) power generation, in order to gauge the net job impacts of renewables and energy efficiency compared to fossil fuels. To reduce any bias that may arise in the 'green jobs' literature with respect to job estimates for fossil-fired technology, one search string combined search terms related to fossil fuel energy with employment creation and job creation metric key words. The policy, quality and skills related search terms were omitted in order to broaden this additional search.

Returned results were filtered manually for relevance to the research questions based on their title and abstract, and each document was given a relevance rating from 1-4 (see Appendix). If this was not sufficient to determine relevance, further inspection of the main text was performed. This paper presents findings from a review of 197 of the most relevant documents (relevance ratings 1 or 2), drawn from a wider body of 704 potentially relevant documents identified through the search strategy.

2.2. How job creation in the energy sector is measured

Methods for estimating the quantity of job creation in the energy sector vary from literature reviews to calculating employment factors[1], collation of data and statistical analysis, to several different types of modelling including statistical (e.g. regression) models, input-output models, or computable general equilibrium (CGE) models and macroeconomic or macroeconometric models. Box 1 summarises several of the most common methods identified in the literature and outlines key advantages and disadvantages. Studies vary in terms of whether they represent direct employment impacts or wider "multiplier" employment created in supply chains or due to increased spending in the economy. Direct employment refers to those jobs that arise directly as a result of an investment; indirect employment commonly refers to the jobs created within the supply chain supporting a specific project. Induced employment can refer to jobs created as a result of the increased household expenditure of direct and indirect employees, but induced effects may also be linked to electricity price and salary changes (Blyth *et al.*, 2014; Stavropoulos and Burger, 2020).

Box 1 Typical methods used to estimate low carbon energy job creation^a

Employment factors and analytical models

Employment factors can be calculated which divide the number of jobs created by a measure of the scale of activity, thereby allowing projects/ programmes of different sizes to be compared, giving an indication of their relative effectiveness in terms of job creation. A common indicator used with respect to green stimulus project / programmes is the number of jobs created per US dollar (USD) (or other currency) invested. Alternative units used for renewable energy specifically are 'jobs per MW capacity installed' or 'jobs per GWh electricity generated'. Employment factors may be derived from industry surveys or interviews, or collated from literature, and then used in a simple, spreadsheet-based analytical model to estimate direct employment impacts.

Economy-wide modelling

Another approach is to take an economy-wide perspective and aim not to count low carbon jobs as such, but to account for wider labour market impacts of green policies. These modelling approaches, which include *input-output analysis* and *computable general equilibrium (CGE) models*, evaluate whether overall jobs or other indicators of the labour market or the economy as a whole are impacted positively or negatively by renewable energy and energy efficiency policies.

Input-Output (IO) analysis requires the use of a set of national IO accounts for an economy. The IO table gives a 'snapshot' of the nature of production and consumption flows in an economy during a specific period of time, usually a year. IO models support evaluation of interdependencies between (i) inputs used and outputs produced by different industries, and (ii) industry outputs and final demand from consumers of goods and services. IO models can be used to estimate the employment impacts of deploying a technology across multiple sectors of the economy. They can quantify the net job impacts of employment gains in one sector (e.g. renewable energy deployment) versus employment losses in another sector (e.g. phasing out fossil fuel generation). IO models can produce separate forecasts of direct, indirect, and induced employment 'multipliers' arising from low carbon energy deployment. In this way, IO analysis can estimate the size of Type 1 (direct + indirect jobs) and Type 2 multipliers (direct + indirect + induced jobs).

A *computable general equilibrium (CGE)* model is an analytically consistent mathematical representation of an economy, which captures the interdependencies across all sectors in the economy at a particular point in time. The models are solved computationally, with an equilibrium being characterised by a set of prices and level of production across all sectors, such that demand equals supply for all commodities simultaneously. CGE models are used to estimate how an economy might react to changes in policy or other exogenous influences, and can capture direct and indirect employment effects, and most types of economy-wide induced effects.

^aThe Box 1 text is based on a range of sources (Blyth et al., 2014; Cameron and Van Der Zwaan, 2015; Yi and Liu, 2015; Mikulić, Bakarić and Slijepčević, 2016; Garrett-Peltier, 2017; Park and Lee, 2017; Mu et al., 2018; Nasirov et al., 2021)

In our review, we compare an international evidence base of identified studies which estimate employment factors for gross job creation relating to different types of renewable energy, energy efficiency, and fossil fuel generation. We collate employment factors from the literature which normalise the quantity of job creation according to the scale of activity, e.g. GW installed or level of investment. The studies reviewed vary in terms of the types of jobs or

employment impacts represented (e.g. direct, indirect and induced) and how boundaries between these categories are defined i.e. which jobs are included as direct or indirect, which induced effects are modelled etc. Another challenge is that there is a high degree of repeat referencing or replication of employment factors across literature focused on the energy sector (Cameron and Van Der Zwaan, 2015; García-García, Carpintero and Buendía, 2020). Our comparative analysis on gross employment factors, presented in Section 3, is therefore indicative. We also review estimates of whether or not low carbon energy transitions might result in net employment gains or losses at country or regional scales. A more limited body of literature identified considers quality and relative skill levels of low carbon energy jobs (e.g. in relation to fossil fuel industries). Such studies may be based upon analyses of employment surveys or occupational profiles derived from input-output tables, and we discuss examples in Section 4.

[1] Defined by Cameron and Van Der Zwaan (2015, p. 161) as “the number of jobs derived from a certain renewable technology investment or capacity”.

3. Review Findings On Quantity Of Job Creation In A Low Carbon Energy Transition

In this section, we review findings and quantitative metrics from the literature as they relate to how many jobs can be created by deploying renewable energy and energy efficiency compared to supporting investment in fossil fuel energy production. We also consider net employment and economic effects of transitioning to a low carbon energy system at a national and regional scale.

In our review, most of the identified studies quantifying job creation by technology estimated gross employment. Therefore, following an approach taken by Blyth et al. (2014), we provide an approximate assessment of net jobs impacts by comparing gross employment factors for renewable energy and energy efficiency with those for fossil fuels. Sections 3.1. and 3.2. review gross employment factors according to different metrics identified in the reviewed literature, relating to the scale of technological activity and level of investment.

Evidence identified in our review on net job creation applies predominantly to international, national and regional scales, and tends to focus more on net effects as a result of low carbon transition across the power sector or wider energy system rather than exclusively as a result of deploying single technologies. In section 3.3 we consider studies which estimate how many jobs are created as a result of decarbonising the energy sector, net of jobs lost in more carbon intensive, fossil fuel activities. These studies attempt to characterise how net job creation, destruction, and linked economic impacts might be manifested at a national or regional level, e.g. as a result of climate change mitigation targets or Nationally Determined Contributions (NDCs). Section 3.4. discusses findings from the literature on the economic impact of low carbon energy job creation.

3.1. Comparing gross job creation of energy technologies and interventions by scale of activity

In Figures 1a and b, we have collated data from our review on job creation per installed capacity for different life stages of electricity generation technologies: manufacturing (in job-years/GW); construction and installation (in job-years/GW); and operation and maintenance (in jobs/GW). One job-year is one full-time job for one person lasting for a year (Dufo-López, Cristóbal-Monreal and Yusta, 2016). These units show the number of jobs created annually and are used to characterise manufacturing and construction/ installation jobs which are required in the first few years of projects and at a project-level tend to be shorter-term in nature compared to ongoing

employment in operation/ maintenance. Operation and maintenance jobs are typically expressed in jobs/MW, as it is assumed that these jobs are more permanent in nature and should last over the lifetime of energy technologies (Ram *et al.*, 2022). These more granular estimates broken down by technology life stage are only available in a limited range of studies in the literature. In Figures 1a and b, we have converted jobs and job-years from the original datasets so they are normalised by GW rather than MW. This helps to aid comparison of the potential volume of jobs created by technology.

Most of the documents from which this dataset has been derived estimate direct and/or indirect employment impacts using an analytical approach (see notes to Figures 1a and b). Two studies apply an input-output model, but only estimates from the global analysis of Jacobson *et al.* (2017) account for direct, indirect and induced jobs. Therefore, the data shown in these figures relates mainly to employment created directly in the activities shown, or in associated supply chains.

A number of key observations can be made with respect to this dataset. Firstly, there is a higher level of manufacturing job creation per GW for several types of renewables (including offshore wind, small hydro and solar PV) compared to gas or coal-fired electricity generation (Figure 1a). The average estimate of job-years created per unit of installed capacity is particularly high for offshore wind (16,800 job-years per GW). The potential to make use of high employment factors for manufacturing depends on the presence of a renewables manufacturing base in any given country. Moreover, many construction/installation and operation/maintenance jobs may effectively be exported overseas depending on the development and size of an export market for manufactured renewables. Simas and Pacca (2014) observe that the standard metric of manufacturing job-years/unit of installed capacity in a particular year may be misleading since it does not account for the proportion of imports or exports. It would therefore be possible for a country to have a very high index if installed capacity is low due to most manufactured technologies being exported.

The evidence on construction and installation (Figure 1a) indicates that this activity creates the most jobs for solar PV, biomass and small hydro, around between 15,000 and 18,000 job-years per GW. Construction of natural gas power plants is associated with the lowest level of employment (2,500 job-years per GW). Wind farm installation performs relatively modestly (averaging 4,200 and 7,200 job-years per GW for onshore wind and offshore wind respectively). Ram *et al.* (2022) observe that demand for construction, installation, and operation and maintenance jobs tends to be created locally, and these activities are therefore a fairly good indication of the potential to generate jobs within a country or region. However, a key uncertainty is the extent to which labour and supply chain services may be imported from other countries.

Figure 1b suggests that operation and maintenance is associated with the highest number of jobs over technology lifetimes for small hydro (1,600 jobs/GW) and biomass (1,100 jobs/GW). Natural gas and coal have the lowest employment factors for operation and maintenance (130 and 155 jobs/GW respectively).

The estimates of job creation presented so far pertain to electricity generation technologies. The review has found only limited examples of equivalent data for heating, low carbon or otherwise. Ram *et al.* (2022) include employment factors for a range of heating technologies in a recent global scenario analysis based on an energy transition to 100% renewables, with a high level of electrification, from 2015 to 2050. The authors present job intensities by manufacturing, construction and installation, and operation and maintenance activities, expressed in units of jobs (or job-years) created per megawatts of thermal (MW_{th}) installed capacity. In general, these particular estimates suggest that CIM activities are associated with a higher number of job-years per megawatts

thermal (MWth) for low carbon heating technologies and fuels sources (e.g. individual heat pumps, district heating sourced from heat pumps or biomass, or energy from waste CHP) compared to gas or oil heating. Estimates for employment creation in operation and maintenance (jobs per MWth) are very similar for individual oil, gas, electric and heat pump technologies, but highest for solar thermal and individual biomass heating systems.

3.2. Comparing gross job creation of energy technologies and interventions by level of investment

Figure 2 summarises the evidence identified across 14 studies which estimate the number of gross jobs created per USD million invested for different energy technologies. This represents total investment, across public and private sectors, and is largely based on studies which model estimates of job creation potential using input-output analysis (see notes to Figure 2). Half of the studies account for direct and indirect employment effects only, with the other half representing indirect and induced employment multipliers as well as direct jobs impacts. Employment factors extracted from five documents reviewed capture investment only or mainly in the CIM phase, with a further four studies quantifying jobs per investment in CIM and O&M phases. However, the remaining six studies are unclear on which phase/s are represented.

The comparison presented in Figure 2 suggests that renewables or energy efficiency can generate more jobs per US dollars invested than fossil fuel generation or nuclear power. Fossil fuel generation creates five jobs per USD million invested on average across the various studies, compared to eight jobs / USD million for nuclear power and 15 jobs / USD million for the renewable energy technologies shown in the chart. Building energy efficiency demonstrates the highest job creation potential per level of investment, creating an average of 22 jobs / USD million.

The relative employment outcomes of investing in higher carbon or low carbon technologies or energy efficiency may reflect a number of factors, including the ratio of spending on local content versus imports, the ratio of spending on labour as opposed to capital, and average wage levels for jobs in particular technology sectors and their supply chains (Pollin and Garrett-Peltier, 2009). The extent to which high employment factors are desirable as a policy objective is a point to which we return later. Lower employment factors for fossil fuel power generation may also be associated with greater market maturity of well established, conventional technologies compared to renewables or energy efficiency.

3.3. Does a shift to low carbon energy create jobs? Estimates of net job creation or destruction at a national or regional scale

Global analyses are in broad agreement that the net employment effects of climate change mitigation policy and a low carbon energy transition will be positive, estimating that by 2030 and 2050, more jobs will be created by a shift to low carbon, renewable energy than the number of jobs displaced from decommissioning fossil-fuel power plants (e.g. (ILO, 2019; Jacobson *et al.*, 2019; IRENA, 2021; Pai *et al.*, 2021). Nevertheless, key questions for policy makers are how job creation and job destruction will be geographically distributed, and how the impacts of this transition will fall in particular countries and regions at different points in time.

In Table 2, we summarise the scope, methods, and findings from 14 studies identified in our review which estimate how decarbonisation in the energy sector affects the overall number of jobs created and displaced at a national and/or regional scale. Taken together, these studies represent evidence from Argentina, Brazil, China, India, Japan and eight European countries (including the UK). Half of the studies in Table 2 focus on the impacts of

decarbonisation in the power sector alone, but several studies consider other energy sectors such as heat, transport and industry (Sievers *et al.*, 2019; Füllemann *et al.*, 2020; Lee *et al.*, 2022). An important variation between the studies relates to how they use counterfactuals to calculate net employment impacts. Across the 14 documents, overall job creation or destruction is typically estimated as being net of avoided investment in fossil fuels (either domestic industries or imports), or net of reference (e.g. continuation of current policy) scenarios. In terms of methods, input-output analysis is most commonly used (five studies) followed by CGE models (three studies), including Perrier and Quirion (2018) who combine both model types. Sievers *et al.* (2019) and Lee *et al.* (2022) apply a macro-economic and macro-econometric model respectively. Five of the identified documents account for direct, indirect and induced employment; five studies consider direct and indirect but not induced jobs; and two studies estimate direct jobs effects alone[2].

The documents reviewed in Table 2 address the net employment impact of climate change mitigation or decarbonisation scenarios over the next one to three decades, achieving higher shares or capacities of renewable energy and/ or substituting fossil fuels with renewables and energy efficiency. Nine of the 14 studies conclude that, at a national scale, there is likely to be positive net job creation overall from replacing fossil fuels with renewables / energy efficiency or as a result of energy sector decarbonisation. Whilst some of these additional jobs are relatively short term since they relate to the construction and installation phase, Arvanitopoulos and Agnolucci (2020) find that in the UK, there is still likely to be an increase in overall jobs in the long term. Perrier and Quirion (2018) observe that both an input-output and CGE model generate positive net employment outcomes in France for a €1 billion investment in solar PV installation or building weatherisation.

Three studies find that a mixture of job creation and displacement effects could result in China, India and Italy respectively (Cai *et al.*, 2017; Mu *et al.*, 2018; Sharma and Banerjee, 2021). Mu *et al.* (2018) model alternative ways of financing a feed-in tariff in China to attain a 1 Terawatt-hour (TWh) expansion target for solar PV and wind power. These lead to direct and indirect job creation but induced job losses, with net job creation arising overall in three scenarios and modest net job destruction in one scenario for wind power compared to a reference scenario.

Two studies report overall net job destruction or indicate a significant risk of such an outcome (Baran, Szpor and Witajewski-Baltvilks, 2020; Le Treut *et al.*, 2021). Le Treut *et al.* (2021)'s CGE model analysis concludes that power sector decarbonisation in Argentina could cause a small proportion of job losses across the economy (-0.5 to -0.7%), and that net job creation in energy, construction and manufacturing would be offset by job destruction in other economic sectors. Additionally, significant structural change would be implied within the energy sector due to the fast decline of fossil fuels and rapid growth of low carbon power generation (Ibid.). Baran, Szpor and Witajewski-Baltvilks (2020) caution that a transition away from coal production and use in Poland could lead to net job destruction, without a well-managed plan to help displaced coal miners transition to new roles in renewable energy and energy efficiency sectors.

The aggregate balance of job creation and destruction across a national economy also depends on the extent to which new and displaced energy sectors utilise labour within or outside a given country. Cai *et al.* (2017) combine input-output and analytical models to evaluate employment impacts of solar PV, onshore wind, hydropower and bioenergy in Italy from 2006 to 2014. The authors contend that lower job creation for renewable energy CIM activities compared to three potential alternative investments in other economic sectors is due to many renewable energy components being imported. Conversely, O&M in Italy's renewables sector used mostly local goods and services and generated higher employment factors than the counterfactual investments. Similarly, in an input-output analysis of a transition to low carbon electricity generation in Japan by 2050, Kuriyama and Abe (2021)

find that a shift to renewable energy could lead to net job creation within the country by reducing the need for fossil fuel imports and reliance on overseas labour. Making the most of this domestic job generating opportunity depends on effective national and local government support providing training, recruiting workers and developing a labour force with requisite skills in renewable energy sectors (Ibid.).

Our review identified limited examples of studies which attempt to quantify the regional distribution of low carbon energy employment. Sharma and Banerjee (2021) find that retiring coal power plants in India and attaining a national 100GW solar PV capacity target would have mixed net jobs impacts across different regions. While the authors' analytical model suggests an outcome of positive net job creation in six Indian states, net job destruction is projected in six other states, particular those which have considerable coal mining activity. Kuriyama and Abe (2021) conclude that transitioning to renewable electricity generation in Japan could be particularly favourable for creating stable, long-term jobs in O&M in rural areas. The study illustrates that early planning is required for conventional power plant phase out to avoid a surplus of workers, accounting for regional differences in the impact of this phase out and new opportunities available for renewables, energy efficiency or other low carbon energy sectors. Elsewhere, Sievers et al. (2019) investigate a low carbon transition scenario across the energy system using a macroeconomic model and an economic impact assessment model; their approach includes a regional breakdown of employment impacts. While the study finds that a low carbon energy transition could create around 1% more jobs in total nationally from 2010 to 2030, northern and eastern German states would likely gain the most economically given their high suitability of locations for siting renewable energy, and that they are less affected by the phase out of fossil fuel power plants.

3.4. Economic impacts of net job creation or destruction at a country or regional scale

The evidence reviewed in Sections 3.1 and 3.2 indicates that renewables and energy efficiency can generate more jobs than fossil fuels for the same level of investment. It does not automatically follow, however, that prioritising investment in these technologies will result in higher employment for any given national economy in the long-term. Investing in options with higher job multipliers such as certain renewable energy CIM activities or installing building energy efficiency interventions makes sense in a depressed economy in which aggregate demand is low compared to potential supply of goods and services (creating a so-called "Keynesian output gap"). In such a context stimulating additional employment in labour-intensive sectors is very likely to lead to higher overall employment (Blyth *et al.*, 2014).

While the circumstances around the COVID-19 economic recession and the 2009 financial crisis are very different, evidence from the 2009 crisis indicates that the green measures (e.g. in renewable energy infrastructure) forming part of the recovery stimulus created more jobs than conventional stimulus measures (Allan *et al.*, 2020). The CCC recommend that in the short term, *"green stimulus policies can be economically advantageous compared to traditional fiscal stimuli. They tend to have higher short run multipliers and higher numbers of jobs created"* (CCC, 2020, p. 141). Domestic construction projects such as insulation retrofits or building wind turbines may be particularly favourable and less prone to offshoring services overseas. In comparison to the renewable energy CIM phase, employment in the operation and maintenance of renewable power generation technologies is typically more permanent, with potential to last over technology lifetimes (Ram *et al.*, 2022).

There is a debate in the literature identified in our review around the extent to which policies supporting renewable energy may contribute to longer term economic growth, notwithstanding short-term employment and growth benefits (e.g. Jaraite, Karimu and Kazukauskas, 2017; Safwat Kabel and Bassim, 2019). While we have not found

extensive evidence to definitively answer this question, identified studies suggest that subject to geographic and contextual variations, low carbon energy shifts may promote modest economic growth effects and are unlikely to be detrimental to economies at a national level. Several of the studies presented in Table 2 suggest that supporting a low carbon energy transition is compatible with longer-term economic growth over the next 10-30 years. For example, Stamopoulos *et al.* (2021) note that increased investments in solar PV and wind power are key to their modelled outcome that the Greek National Energy and Climate Plan could contribute EUR 6.8 billion to Greek GDP by 2030. Similarly, Sievers *et al.* (2019) project a 1.6% increase in GDP versus a reference scenario in Germany from 2010 to 2030 as a result of energy sector decarbonisation. Elsewhere, Lee *et al.* (2022) analyse two Net Zero policy scenarios for Japan (with and without nuclear power phase out), finding that by 2050 decarbonisation across the energy system could add 4%-4.5% to GDP compared to a reference scenario.

On the other hand, and linked to the challenge of allocating displaced coal miners to new low carbon energy jobs, there is a significant risk that replacing coal power with renewables and energy efficiency in Poland could lead to reductions in GDP as well as net labour losses (Baran, Szpor and Witajewski-Baltvilks, 2020). The contribution of different renewable energy phases to net employment and value added in a given country may also depend upon the extent to which goods, services and labour are sourced locally. Cai *et al.* (2017) find that due to significant component imports, renewable energy CIM in Italy contributed less to value added historically than counterfactual scenarios. By contrast, the predominantly local content and higher labour intensities of renewables O&M led to higher value added than the counterfactuals. Pegels and Lütkenhorst (2014, p.529) found that renewables component manufacturing in Germany was “*often located in the traditional industrial centres*”, some with the “*highest unemployment ratios nationwide*”.

While well established, high carbon technologies and sectors may be close to their limits in terms of additional innovation and economic productivity gains, investment in less mature, faster growing low-carbon technologies such as renewables could contribute more to productivity through greater scope for innovation and learning by doing (CCC, 2019). Careful planning is required to optimise the design of subsidy schemes for renewable energy technologies to lower risks of increasing income inequality or regressive distributional impacts on low-income households, such as those which have been modelled in studies of Ireland and Germany respectively (Többen, 2017; Farrell, O'Donoghue and Morrissey, 2020). When designing stimulus programmes, it makes sense from a Keynesian perspective to support technologies and projects that have a positive, long-term impact on economic productivity beyond the timeframe of the direct stimulus effects. In a longer-term context, labour intensity is not in and of itself economically advantageous. If it implies lower levels of labour productivity (economic output per worker), then it could adversely affect prospects for long-term economic growth. On the other hand, critiques of Green New Deal approaches argue that they perpetuate a narrative around the need for full employment, high production, high consumption economies which are incompatible with conserving finite resources and minimising ecological footprint (Green, 2022). Developing on from Blyth *et al.* (2014), we therefore contend that policy should not be focused on maximising jobs per unit of investment in the long run. Rather, policy decisions should be based on whether investments can contribute to an economically efficient transition towards effective climate change mitigation, taking account of the need for a just transition, wider ecological impacts and energy security considerations.

[2] Two documents do not provide information on the representation of direct, indirect and/or induced jobs.

4. Review Findings On Quality Of Job Creation, Occupations And Skills In A Low Carbon Energy Transition

Mattos (2018) suggests that “*evidence on green policy impacts on job quality is minimal.*” In our review of international literature, we have identified a subset of studies which discuss the quality of green or low carbon energy jobs in qualitative terms. Lambert and Silva (2012) have previously noted that developing job ratios to characterise job type or quality is more challenging than counting the number of jobs created. We have found a limited number of studies which use quantitative metrics to assess employment quality, occupation types or relative skill levels in low carbon energy compared to high carbon energy sectors. In this section, we discuss key highlights from relevant literature related to these topics.

4.1. Quality of job creation

The Paris Agreement (UNFCCC, 2015) sets out that national priorities for low carbon development should ensure a just transition through the creation of decent work and quality jobs. Several studies extracted from our review highlight the importance of ensuring the quality of green jobs, and some documents refer to the concept of ‘decent work’. For example, Mattos (2018) writes that “*green jobs are, by definition, decent jobs, i.e. a subset of jobs in environmental sectors which provide adequate wages, safe working conditions, safeguard workers’ rights and social dialogue, and which provide social protection.*” In other documents identified in our review, higher job quality is described for example in terms of high wages and full-time employment (Jung, 2015), and permanent rather than temporary jobs (MacCallum, 2016; Mattos, 2018).

However, green or low carbon energy jobs may not always or necessarily be ‘higher’ quality jobs. It has been suggested variously in the literature that direct employment in renewable energy construction or installation may be linked to temporary work which expires on completion of specific projects or might no longer be needed once renewable energy capacity targets have been met (MacCallum, 2016; Sofroniou and Anderson, 2021; Godinho, 2022). Similarly, the UK’s smart meter installation programme will create the need for smart meter installers as a new occupation – but the longer-term job security or career prospects of these jobs requires further exploration (Sofroniou and Anderson, 2021).

With reference to data on Scotland, Connolly, Allan and McIntyre (2016) suggest that labour intensity may fall as renewables mature and employment needs shift from construction to maintenance and servicing. The authors observed that “*between 2007 and 2012 the number of LCEGS [Low Carbon Environmental Goods and Services] jobs declined whereas the installed capacity of renewable generation in Scotland more than doubled*” (Connolly, Allan and McIntyre, 2016). Nevertheless, it is likely that achieving national net zero or decarbonisation targets will require a prolonged period of construction and installation of new renewables capacity over the next several decades.

Godinho (2022) point to OECD research which indicates that in general, lower skilled employees in the energy supply sector are more likely to be affected by job displacement as a result of low carbon energy transitions, and this may be compounded by workers losing benefits from previous work in carbon-intensive sectors (Chateau, Bibas and Lanzi, 2018; Botta, 2019). In general, stronger national and international labour market standards and regulations are needed to support unemployed workers in decommissioned industries, provide skills training and increase the availability of alternative, decent work in affected regions (IRENA and ILO, 2021; Godinho, 2022). We consider the issue of skills in the following section.

4.2. Skills and occupational profiles

A cluster of studies was identified around the issue of skills and typical education levels associated with green jobs. Based on an analysis of US employment and occupational data, Consoli *et al.* (2016) suggest that *“Green jobs exhibit higher levels of education, work experience and job training”* and *“use more intensively high-level cognitive and interpersonal skills compared to non-green jobs”*. This is supported by the findings of a study (Elliott and Lindley, 2017) that analysed data from the US Bureau of Labour Statistics, and found that green industries *“increased the quantity of workers demanded from the middle of the skill distribution at the same time as they reduced the quantity demanded for lower skilled workers”*. The same study went on to conclude that it was *“College graduates who gain the most from the expansion of green jobs”*. A German study (Pegels and Lütkenhorst, 2014) used data from the Federal Ministry for the Environment to assess the costs and benefits of Germany’s Energiewende, focussing on wind and solar PV. The study found that *“the share of university-degree staff is around three times as high as the national industry average”*.

Allan *et al.* (2021) classify jobs in the UK offshore wind sector according to skill level, through a method involving extraction of data from a UK input-output (IO) table. The data is from 2010, the latest available to allow a detailed breakdown by occupational skills. The authors aggregate nine occupational SIC (Standard Industrial Classification of economic activities) categories to form three groups representing high, medium and low skill levels. Figure 3 indicates that almost 90% of offshore wind jobs in 2010 were in the high to medium skill categories, and that most jobs are created indirectly in the supply chain or through the induced effect of additional household expenditure.

Vona, Marin and Consoli (2018) also report high skill levels and salaries for green occupations compared to non-green employment in the US. Despite some identified differences in skills levels between green and non-green jobs, Reddy (2016) suggests that the skills required are not new, and Bowen, Kuralbayeva and Tipoe (2018) propose that much retraining in the green economy can occur ‘on the job’. Others contend that the greening of the economy will require new skills, competencies and qualifications, linked to the creation of new markets and activities (Aceleanu, Serban and Burghilea, 2015; Shanghi and Sharma, 2014). Several studies note the potential to train and employ young people in these new areas, while helping to address youth unemployment (Aceleanu, Serban and Burghilea, 2015; Rutkowska-Podołowska, Sulich and Szczygieł, 2016; Sulich, Rutkowska and Popławski, 2020). Kapetanious and McIvor (2020) highlight that in the UK, younger and male workers are concentrated in brown sectors, where most jobs require low- or medium-level skills. This implies challenges for younger males working in routine or manual brown occupations if they need to switch to green jobs; they may have to learn new skills in order to avoid job displacement difficulties (Kapetanious and McIvor, 2020).

The identified evidence does not support a straightforward claim that renewable energy or energy efficiency jobs are more skilled than higher carbon energy sectors, and this depends on the extent to which direct, indirect and induced job creation are considered. Several analyses (e.g. Allan and Ross, 2019; Dominish *et al.*, 2019) observe that significant shares of employment in fossil fuel generation and extraction are in higher-skilled occupational categories. There is also demand for lower-skilled, manual occupations which may for example comprise much of solar PV installation and offshore wind construction activities.

Figure 4 presents a chart of data sourced from Dominish *et al.* (2019), which compares the share of occupation types and categories for several renewable electricity generation technologies with fossil fuel power generation (and associated activities). The data for solar PV and onshore and offshore wind has been drawn from

international surveys carried out by IRENA (2017a, 2017b, 2018). This has then been compared to equivalent occupational data derived from the Australian 2016 national census, which has been scaled using regional employment multipliers to represent different world regions (Dominish *et al.*, 2019).

The relative distribution of occupations shown in Figure 4 highlights areas where low skilled jobs are particularly concentrated: for example, assemblers comprise the majority of jobs in solar PV installation, and approximately half of all jobs in coal mining are in elementary occupations. Around 70% of offshore wind construction jobs are comprised of ship crew. Plant and machine operators, assemblers and elementary occupations contribute a small but significant share of wind manufacturing and construction activities. Whilst it is clear that higher-skilled professional occupations comprise substantial shares of operation and maintenance jobs in wind and solar PV, Figure 4 does not indicate clear differences between renewables and fossil fuels in terms of relative skill levels.

The evidence reviewed suggests that green skills supply and demand should be carefully managed through policies supporting green job creation, and coordination of training activities. According to OECD/Cedefop (2014, p.12) this *“will prevent the situation of green skills demand being stimulated by government policy, but not being matched by equivalent action to meet this demand, leading to skills bottlenecks and/or programme failure due to unskilled operators.”* In the context of India, Reddy (2016, p.300) suggests a need for *“collaboration between government authorities and business houses to develop industry-endorsed training programs that give graduates nationally recognised technological skills and provides skilled employees with a diploma certificate. Secondly, there is need to create a nationwide, online skill database that would link students, colleges and employers.”*

In order to facilitate a just transition to a low carbon energy system, IRENA and ILO (2021) emphasise the need for proactive policy which sets out to anticipate and plan for a series of misalignments that are likely to occur. Each type of misalignment is reflected in a number of studies discussed in our paper. These include: temporal misalignments between the pace and scale of job losses and the rate and capacity of job gains required to compensate them; geographic misalignments between the location of new jobs and the regions in which displaced workers live; skills and job role misalignments between outgoing and incoming energy industries; and value chain misalignments as transitions away from fossil fuels create a shift from conventional mining and fuel extraction to sourcing of renewable energy components and materials (Ibid.).

5. Conclusions

In this paper we present findings from a systematic literature review on job creation, quality, and skills in the energy sector, focusing on a shift to renewable energy and energy efficiency. The international literature has revealed that various methods and units are used to estimate the quantity of low carbon energy job creation. In general, much greater standardisation of methods would be desirable in order to compare how many jobs could be created by transitioning to low carbon energy and deploying energy efficiency at a range of scales – from individual projects, through to regional, national and global. Our findings also underline a relative lack of metrics and data in the literature on job quality, skills, and the geographic distribution of employment impacts in decarbonising energy systems, and these should be priority areas for further research.

We compare a range of recent gross job creation estimates which indicate that overall, investment in renewable energy and energy efficiency can deliver more jobs than gas or coal power generation. This finding is consistently supported across a range of different job creation metrics and when focusing on different technology life stages, i.e. manufacturing, construction and installation, operation and maintenance. This suggests that policies

supporting renewables and energy efficiency may lead to net job creation compared to the counterfactual of jobs which may otherwise have been created by investing in fossil fuels. Gross jobs per USD million invested are found to be highest on average for wind power and building energy efficiency interventions, across 14 studies reviewed. Caution should be applied to interpreting data on gross employment per level of investment for any given technology, which may be indicative of various factors including relative technological maturity, share of spending on local content or imports, share of spending on labour (versus capital) and average salary levels.

We identify 14 recent additional studies which estimate the net employment effects of decarbonising part or all of the energy sector at a national scale. These national studies largely agree that the most likely outcome over the next few decades is a modest net positive creation of jobs. This implies that the number of jobs created in renewables and energy efficiency sectors will outweigh the number lost as conventional fossil fuel power plants are retired.

Several studies illustrate how overall net job creation in a given country may reflect a balance of net jobs gains and losses in different regions, economic sectors and parts of the value chain. This balance also depends on the nature of the value chain in outgoing and incoming energy industries and how much a country may rely on imports for fossil fuels, manufacturing of renewables components or labour based in other countries (or indeed, the extent to which there is an export market for domestically manufactured components). In certain regions, jobs in fossil fuel industries may be lost faster than the pace at which low carbon energy sectors can offer new employment. There may be mismatches between regions where displaced workers live and where new opportunities become available, which may be a barrier to accepting alternative employment even if former workers have the requisite skills. In these cases national government transition plans are recommended, co-ordinated with local governments, to manage the impacts of displacement from carbon intensive sectors and respond to the need to build a new low carbon workforce including through skills development and training. Overall, there appears to be limited evidence in the literature reviewed on the geographic and sectoral implications of low carbon energy job creation, and more research is required to improve understanding of issues around local or regional displacement and substitution of employment in high carbon sectors.

Given the current context of high energy prices and consolidating economic recovery from COVID-19, we identify literature on the extent to which renewable energy and energy efficiency support policies may contribute to short-term and longer-term economic growth. This literature is not extensive but indicates that implementing a low carbon energy transition at a country scale could lead to moderate economic growth as well as positive net job creation overall. However, as with employment effects, this could mask regional and sectoral disparities and possible negative economic impacts in regions with a substantial presence of high carbon industries. Policies supporting domestically-based manufacturing and installation of renewable energy, and building energy efficiency retrofitting, may be particularly effective at creating short-term jobs and economic stimulus. In comparison, operation and maintenance jobs can potentially last over energy technology lifetimes.

Jobs created per unit of investment represent only one aspect of a low carbon transition; what matters in the longer-term is whether the investment contributes to an economically efficient transition towards a country's strategic goals, considering environmental impacts, the need for a just transition and energy security. Meeting decarbonisation and net zero targets internationally implies a continuous need for jobs over several decades, e.g. to build the new renewables capacity needed for likely greater demands for electrification from transport, heating and cooling, and to carry out widespread low carbon and energy efficiency retrofits across national building stocks. Wider impacts of such activities go well beyond job creation to include co-benefits such as improved air

quality, more comfortable homes, more resilient energy supplies and reduced dependence on fossil fuels, whether they are sourced domestically or via imports.

There is a substantial literature on the quantity of job creation that may arise from decarbonisation in the energy sector. By comparison, we found a more limited subset of studies which investigate implications of such a transition for employment quality or relative skill levels. It is desirable that a low carbon energy transition should create quality jobs, for example in terms of adequate wages and employee rights, full-time employment, safe working conditions, and permanent rather than temporary jobs. Direct employment in renewable energy manufacturing, construction or installation has been linked to temporary or short-term work in several documents reviewed. It has been suggested that such employment is likely to expire on completion of specific projects. Sequential planning will be required to train and coordinate local workforces required for renewables expansion and building retrofits, minimising time gaps between projects and the need for workers to relocate.

There is agreement in the identified literature that green jobs in general tend to be more highly skilled compared to higher carbon occupations. However, it would be overly simplistic to suggest that renewable energy or energy efficiency jobs are necessarily more skilled than employment in fossil fuel sectors. The majority of jobs in the operation and maintenance of wind power and solar PV are in highly skilled, professional occupations. There is also demand for lower-skilled, manual occupations which for example comprise significant shares of solar PV installation and offshore wind construction activities. Nevertheless, several studies suggest that lower-skilled workers are most likely to be affected from job losses as a result of low carbon energy transitions. Stronger international and national labour market standards and regulations are required to ensure displaced employees have access to alternative, decent work in low carbon sectors. There is a need for national policy makers to coordinate the development and supply of training so that it takes full account of the wide range of occupational functions required for manufacturing, installing and servicing renewable energy and energy efficiency technologies.

Declarations

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Tables

Table 1 Search terms applied in systematic review

Employment and job creation	Energy and environment	Fossil fuels / nuclear	Job creation metrics	Policy / techno-economics	Geography, quality and skills
job	energy	gas	jobs/MW	policy	local
"green job"	green	"natural gas"	jobs/GW	"net zero"	regional
"low carbon job"	renewable	"shale gas"	direct	subsidy	location
employment	"low carbon"	coal	indirect	incentive	geography
"employment creation"	decarbonisation	fossil	induced	stimulus	quality
"job creation"	"renewable energy"	"fossil fuel"	multiplier	invest	skill
"job destruction"	"energy efficiency"	nuclear	"employment factor"	spend	qualification
net	climate change		"supply chain"	CAPEX	"decent work"
				OPEX	"just transition"

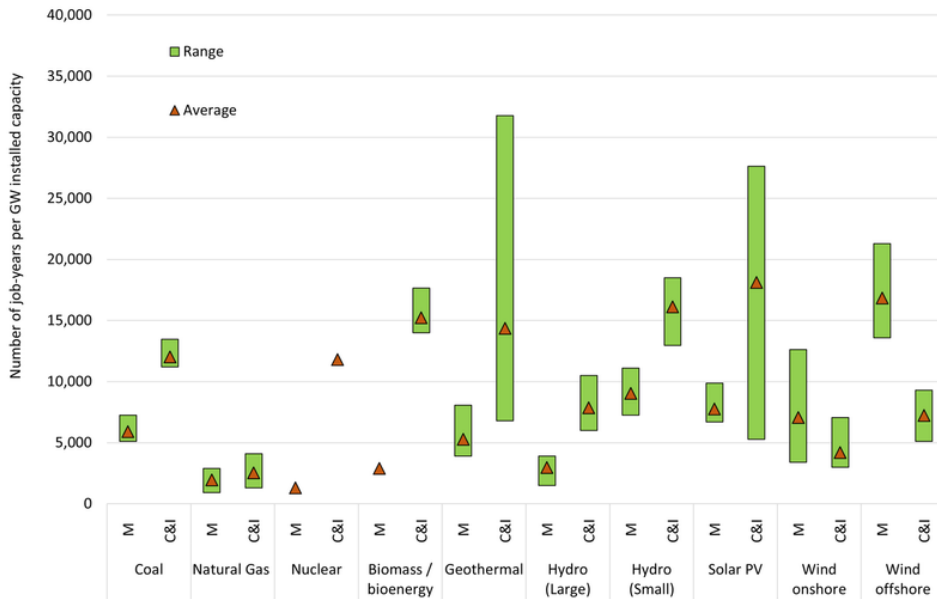
Table 2 Net job creation / destruction and economic impact from supporting renewables and energy efficiency: national and regional scale estimates

Author (Year)	Geography (Timescale)	Energy sectors included	Method (Direct / indirect / induced jobs included?)	Findings on net job creation / destruction and economic impact
Le Treut <i>et al.</i> (2021)	Argentina (2015-2050)	Power sector decarbonisation.	Deep decarbonisation pathways developed using the LEAP model and loaded into IMACLIM-Country - a multi-sector CGE model (Direct, indirect and induced jobs).	Net job creation is slightly negative across the whole economy (-0.5 to -0.7%) reflecting sectoral net job creation in energy, construction, and manufacturing, but net job destruction in agriculture, transport, and services.
Heideier <i>et al.</i> (2020)	Brazil (2017-2040)	Solar PV distributed generation (PVDG) and energy efficiency - impacts of wider diffusion on markets of three power distribution utilities.	Accounting simulation model scenarios (Direct jobs).	Net job creation (gross jobs created less a counterfactual of jobs created by investment in thermal power plants with equivalent electricity output) was positive for PVDG and energy efficiency measures for all three distribution utilities.
Mu <i>et al.</i> (2018)	China (based on 2012 Input-Output table).	Effect of expanding solar PV and wind power by 1TWh under scenarios with alternative renewable energy financing mechanisms, compared to 2012 reference scenario.	CGE modelling: China Hybrid Energy and Economic Research model (Direct, indirect and induced jobs).	Direct and indirect effects lead to job creation and induced effects cause job losses. Overall, 3 of 4 scenarios for solar PV and wind power reveal net job creation of 13,600 to 34,100 jobs/TWh, and 1 scenario net job losses of 800 jobs/TWh.
Perrier and Quirion (2018)	France (2013)	€1 billion investment in solar PV installation or building weatherisation compared with €1 billion avoided investment in power, gas and heat sectors.	Input-output and CGE modelling (N/A)	Solar PV installation and building weatherisation have positive net employment impacts across both IO and CGE models applied, but positive impacts are higher in the IO model.
Sievers <i>et al.</i> (2019)	Germany (2010-2030).	Power sector, buildings heating and transport, industry, renewable energy and energy efficiency, fossil fuels.	Low carbon energy transition scenario and macroeconomic / economic impact assessment models of national economy, regional and sectoral distribution of	By 2030, energy transition scenario results in: 1.1% higher total employment nationally and 1.6% higher GDP compared to reference scenario (no new policies). Northern and eastern German states may benefit the most economically as they have attractive locations for renewable energy and are less

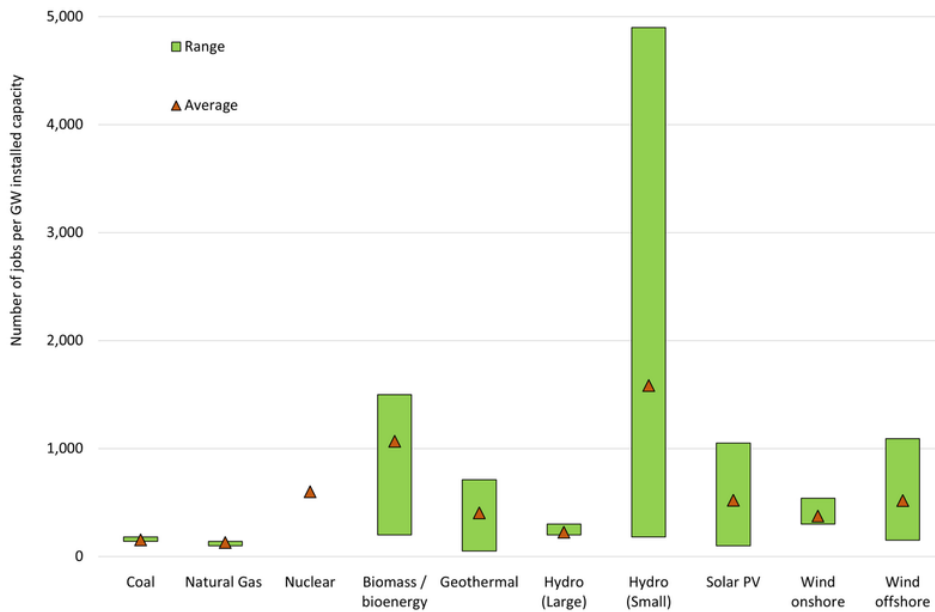
			economic and employment impacts (Direct, indirect and induced jobs).	impacted by declining fossil fuel power generation.
Stamopoulos <i>et al.</i> (2021)	Greece (2020-2030)	Transition from fossil fuels to renewable energy for electricity production including phase out of lignite power generation by 2028.	Input-output modelling to assess impact of increased investment in a higher share of renewables under the Greek National Energy and Climate Plan (Direct, indirect and induced jobs).	Estimated that implementing Greek National Energy and Climate Plan would create 154,000 jobs from 2020-2030 and contribute EUR 6.83 billion to Greek GDP. This overall positive economic impact is attributed mainly to investments in solar PV and wind generation.
Sharma and Banerjee (2021)	India and regional states (2017-2022)	Power sector, specifically: solar PV; land use change resulting from siting new solar power plants; coal power stations and coal mining.	Analytical, model using employment factors to assess geographic distribution of job creation / destruction impacts (Direct and indirect jobs).	Achieving national target of installing 100GW of solar PV by 2022 and retirement of coal power stations would lead to net job creation in six southern and western states but net job destruction in six other states with significant coal mining activity, particularly in four states in eastern India.
Kamidelivand <i>et al.</i> (2018)	Ireland (based on 2010 input-output table)	Energy and non-energy sectors; substituting electricity from imported gas and coal with renewables.	Input-output substitution model (Direct and indirect jobs).	Substituting gas with renewables creates 26.2 net jobs for each 1 per cent GHG reduced.
Cai <i>et al.</i> (2017)	Italy (2006-2014)	Solar PV, onshore wind, hydropower and bioenergy contribution to value added and employment.	Input-output analysis combined with analytical model accounting for imports (Direct and indirect jobs).	Renewable energy CIM job creation and value added lower than three counterfactual scenarios - attributed to significant import shares of renewable energy components. Renewable O&M phase used mainly local goods and services and had generally higher labour intensities and value added than counterfactual activities.
Kuriyama and Abe (2021)	Japan and regions (2016-2050)	Power sector decarbonisation using renewable energy.	Employment factors based on input-output tables, and Just Transition scenarios for power sector decarbonisation (Direct and indirect jobs).	Shift to low carbon electricity generation via renewables leads to net job creation domestically, particularly in O&M and stable, long duration employment in rural areas. Reduced fossil fuel imports from abroad implies a significant decline in overseas labour.
Lee <i>et al.</i> (2022)	Japan (2018-2050)	Power generation, transport, building heating, steel and other	E3ME macro-econometric model and two policy scenarios (including carbon	Net Zero policy scenarios lead to increased GDP (4%-4.5%) and employment (2%) by 2050 compared to reference scenario (based on continuation of

		industrial sectors.	tax) to reach Net Zero by 2050 (Direct, indirect and induced jobs).	current policies and 32% CO ₂ emissions reduction by 2050).
Baran, Szpor and Witajewski-Baltvilks (2020)	Poland (2020-2030)	Substituting coal with renewables, energy efficiency or natural gas imports.	Dynamic stochastic general equilibrium model (N/A).	Net loss of labour and GDP reductions would result unless former coal miners can move to jobs in green or "neutral" sectors.
Fülleemann <i>et al.</i> (2020)	Switzerland (2011 / 2050)	Switching from fossil fuels to renewable electricity generation, greater energy efficiency in buildings, industry, transport and services.	National energy transition scenarios and input-output tables (Direct and indirect jobs).	Net 20% increase in FTE jobs (particularly in heating and building retrofit).
Arvanitopoulos and Agnolucci (2020)	UK (2030)	Power generation sector: conventional thermal, CCGT, nuclear, renewables.	Econometric: Vector error correction model and UK Times model decarbonisation scenarios (Direct jobs).	Five alternative decarbonisation scenarios result in net employment creation ranging from 12,000 to 152,500 jobs by 2030.

Figures



(a)



(b)

Figure 1

a Gross job-years created in manufacturing (M) and construction and installation (C&I) per GW of installed capacity

b Gross jobs created in operation and maintenance per GW of installed capacity

Notes to Figures 1a & b

a. Data reflects average and minimum / maximum values extracted from a range of studies (Ortega *et al.*, 2015, 2020; Atilgan and Azapagic, 2016; Henriques, Coelho and Cassidy, 2016; Jacobson *et al.*, 2017; Dominish *et al.*,

2019; Ram, Aghahosseini and Breyer, 2020; Ram *et al.*, 2022).

b. The job creation data relates to the global scale, Europe, Portugal and Turkey.

c. Methods: six studies derive employment factors from literature, of which five studies utilise analytical methods; two studies use input-output analysis.

d. Multipliers: four studies include direct jobs only; three include direct and indirect jobs; one study includes direct, indirect and induced jobs.

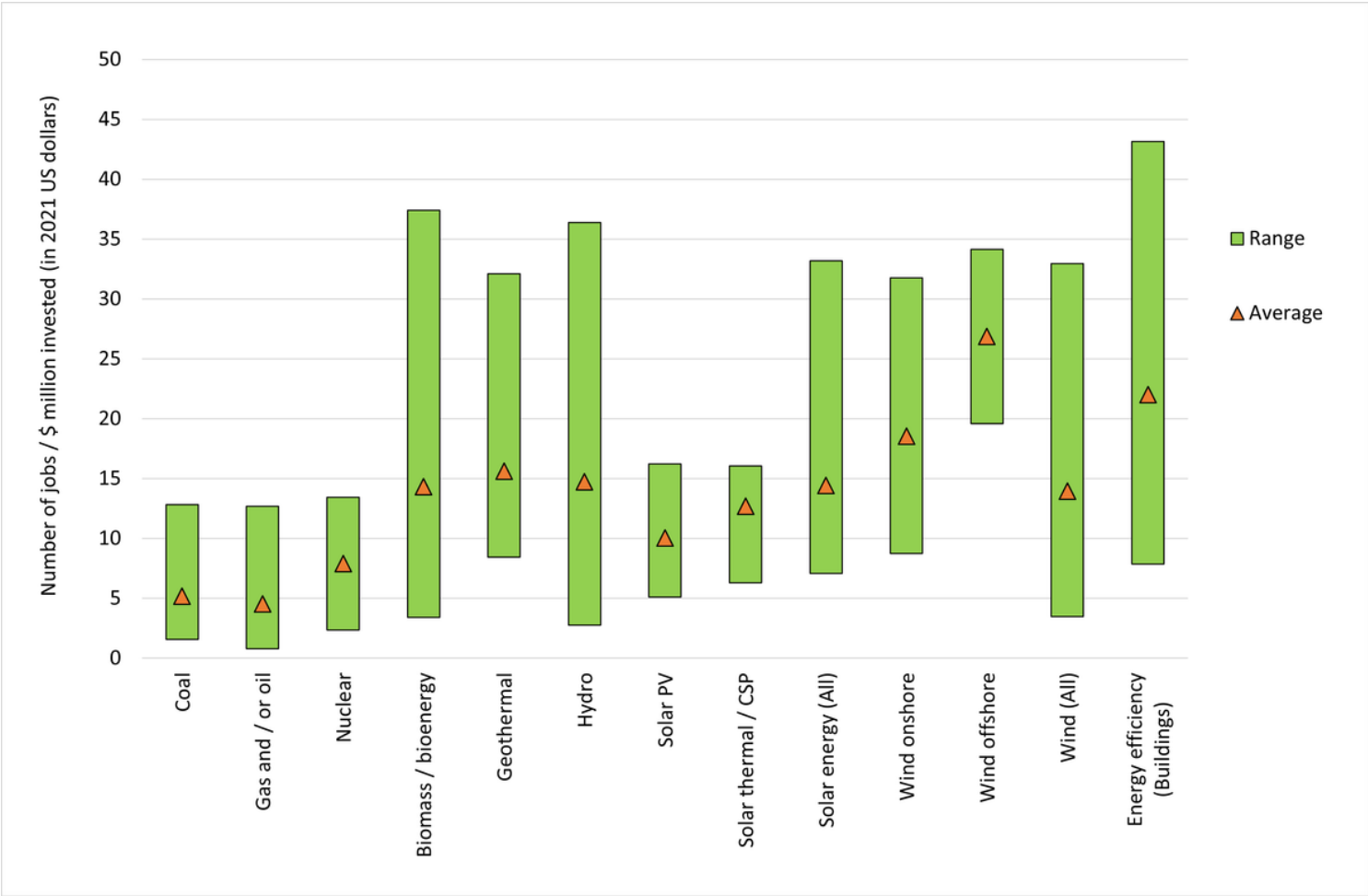


Figure 2

Gross jobs created per USD million invested

Notes to Figure 2

- a. This chart shows identified evidence from 14 studies on number of gross jobs created per USD million (US dollars) invested for different energy technologies or interventions.
- b. The 14 studies were published from 2009 to 2020, pertaining to Europe, Croatia, Germany, Greece, Spain, UK, US, and India (Pollin and Garrett-Peltier, 2009; Lambert and Silva, 2012; Zabin and Scott, 2013; Markaki *et al.*, 2013; Pollin *et al.*, 2014; Rosenow, Platt and Demurtas, 2014; Calzadilla *et al.*, 2014; Mirasgedis *et al.*, 2014; Cambridge

Econometrics, 2015; Mundaca and Luth Richter, 2015; Reddy, 2016; Mikulić, Bakarić and Slijepčević, 2016; Garrett-Peltier, 2017; Brown, Soni and Li, 2020).

c. *Methods:* Nine studies use input-output analysis, including two which also use an analytical method and one which also employs a global CGE model. Two studies employ alternative model types. Three studies collate data from different literature sources.

d. *Multipliers:* Seven studies include direct, indirect and induced jobs; seven studies account for direct and indirect jobs only.

e. *Phase:* Five studies include only or predominantly the CIM phase; four studies include both CIM and O&M phases, and six documents are not clear on their inclusion.

f. *The investment currency in the original datasets has been converted from EUR or pound sterling to USD, based on Eurostat (2023) and OECD (2023), and then adjusted for inflation to 2021 USD using an inflation calculator (U.S. Bureau of Labor Statistics, 2023).*

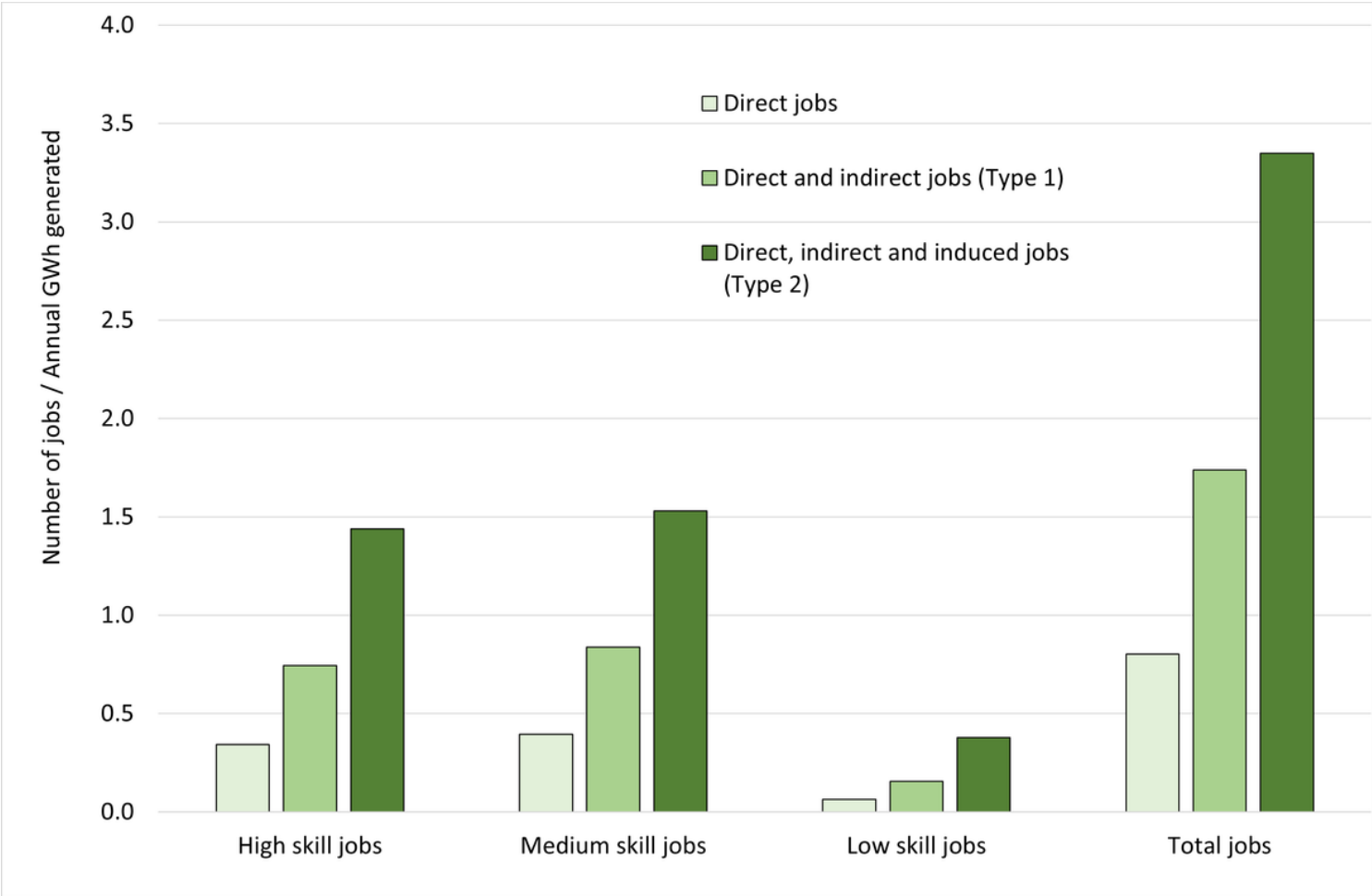


Figure 3

Offshore wind in the UK: Gross jobs per GWh by skill level, direct, indirect, and induced jobs

Notes to Figure 3

a. Data derived from Allan et al. (2021).

b. High skill jobs - Managers, directors and senior officials; professional; associate professional and technical occupations.

c. Medium skill jobs - Administrative and secretarial; skilled trades occupations; caring, leisure and other services.

d. Low skill jobs - Sales and customer services; process, plant and machine operatives; elementary occupations.

e. The factor by which indirect or induced jobs increase for a given increase in direct jobs is the multiplier, Indirect job multipliers are often referred to as 'type 1', while induced job multipliers (including direct, indirect, and induced jobs) are referred to as 'type 2'.

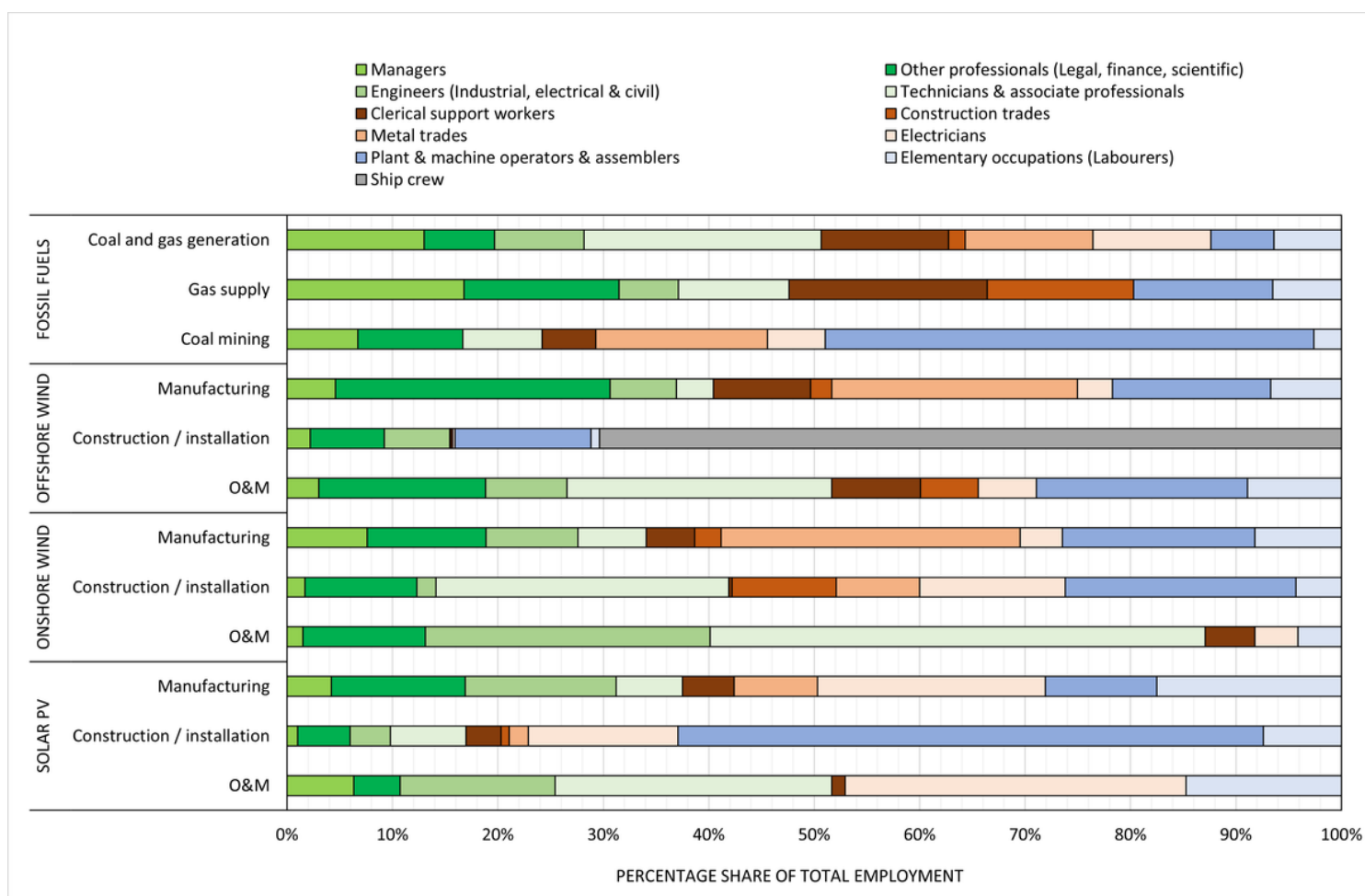


Figure 4

Share of occupational categories by electricity generation technology / fuel source and activity

Notes to Figure 4

a. Data sourced from Dominish et al. (2019).

b. The occupational data is based on surveys of wind and solar PV industries in various developed and developing countries carried out by the International Renewable Energy Agency (IRENA, 2017a, 2017b, 2018).

c. Occupational data for fossil fuels has been derived from labour statistics in the Australian 2016 national census, and adjusted using regional job multipliers to account for different labour intensities in different parts of the world (Dominish et al., 2019).

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- [Appendix.docx](#)